boundline

Borrow against your stocks on Base. Paid in your currency.
Never liquidated while Wall Street is closed.

The session-aware credit layer for Coinbase's tokenized stocks on Base: an onchain market calendar, oracles that refuse a frozen price, protection that runs on weekends, and credit lines any wallet or broker can embed. Non-custodial.

Farseen Shaikh · TODO: second developer · Colosseum Crypto World's Fair, Base track · October 2026

The stocks trade all weekend. Their price does not.

Feed	Held print	Age at reopen	Rounds while closed	First print	Gap
AAPLc	336.84	52.5 h	0	+51 s	–5.8 bps
AMZNc	261.80	52.7 h	0	+41 s	+51.9 bps
GOOGLc	352.63	54.7 h	0	+31 s	+4.7 bps
METAc	719.92	54.4 h	0	+53 s	+14.7 bps
MSFTc	536.08	53.3 h	0	+23 s	–1.5 bps
MSTRc	154.61	53.2 h	0	+21 s	+201.0 bps
NVDAc	230.26	56.7 h	0	+23 s	+1.1 bps
SNDKc	1,581.77	52.4 h	0	+21 s	+144.7 bps
SPCXc	162.62	52.1 h	0	+37 s	+94.9 bps
TSLAc	383.97	55.9 h	0	+23 s	+16.5 bps

- The Chainlink feeds for Coinbase's tokenized stocks publish **Sun 20:00 → Fri 20:00 ET** and hold Friday's close in between.
- Aave V4′s Equities Hub and Morpho's B20 markets read that held price with **no staleness, pause or market-hours check**. Liquidation is open at any hour.
- At the reopen the weekend's move prints in **one block**. A position that gapped through its limit is gone before its owner can act.

Base's docs: "never settle or liquidate against a frozen feed". Aave's risk provider calls session-aware handling "a future stage".

The 9–11 Oct 2026 weekend, all ten feeds: captured live by the Observatory and reconstructed from the Chainlink round history on Base mainnet, agreeing on every field (`docs/evidence/reopen-2026-10-11.md`). 2–4 Oct looked the same: zero rounds, first prints 23–53 s, MSTRc +282 bps.

WHAT IT IS

A credit layer that knows when Wall Street is closed.

Refuse

a frozen price at the venue: on Boundline's markets nobody can borrow or liquidate against it; repay and add-collateral always work.

Protect

on Saturday, at the price the market actually shows: a small sale paid for by the collateral, or a repay from a reserve.

Embed

a line of credit any wallet or broker can offer, paid out in the user's own currency, with one record per loan.

- **Who pays:** wallets, neobrokers and fintechs that list tokenized stocks (B2B2C); curators who want protected markets; issuers who need credit rails.
- **Non-custodial:** the user keeps their own Morpho or Aave position and authorises Boundline, or the venue's own audited contract, with one signature.

HOW IT IS BUILT

From the clock to the partner: every layer is live.

Layer	What decides	Proof
Time	`MarketCalendar`: every NYSE session onchain (pre, regular, post, overnight, weekend, holiday, early closes, daylight saving)	5,643 instants, 3 independent implementations, 0 disagreements
Verdict	`SessionGate`: per stock, is this price safe? Session, feed age, Coinbase OracleRegistry pause, issuer transfer pause; fails closed	34 oracle tests
Venue oracle	`SessionGuard` (Morpho): reverts while the feed is held, so borrow, withdraw and liquidate stop; repay never reads it	live since 3 Oct; mainnet fork
Protection	`ShadowGuard`: the 24/7 Aerodrome TWAP less a haircut. Morpho's audited pre-liquidation + Boundline's `Protector`; the `DeskProtectionVault` ladder and the `BoundlineReserve` (ERC-4626)	keeper acted on its own, also under a frozen feed; Reserve advances settled
Credit	`CreditDesk` (Morpho), `AaveDesk` (Aave V4 via Taker/Giver): partner policy, decision hashes, local-money payout	live lines; Aave on a mainnet fork
Data	Observatory, indexer with hash-chained daily roots, fork replay through weekend gaps	live on mainnet reads
Partners	API (x402-metered), SDK, app, console, MCP agent surface	eight Workers live

ONE WEEKEND, PROTECTED

Repay is always open. Sales wait for the session.

Fri 20:00 ET
The feed freezes. `SessionGuard` stops pricing it: no borrows, no liquidations on Boundline's markets. Repay and add-collateral stay open.

Saturday
The token moves on Aerodrome. `ShadowGuard` prices the position at the TWAP less 2%. The keeper sizes the action **on chain**.

Saturday, inside the band
A slice of collateral is sold inside Morpho's pre-liquidation callback and repays the debt; or the vault repays from the user's reserve, then a bounded Reserve advance.

Sun 20:00 ET
The feed reopens. The position is already safe. Any advance is settled in session, at a fresh price, with its fee.

Why this order is forced: Morpho's pre-liquidation refuses once a position is past its limit, so nothing can rescue a position that gaps through it at the reopen. Only an action taken **while the feed is frozen**, at the live price, prevents the liquidation. The venues cannot see the live price; Boundline does.

Base itself holds the keeper's action until the feed state matches.

- Each protective action is signed with a **storage predicate** on the feed's `updatedAt` slot and a block bound, sent with `base_sendRawTransactionValidity`.
- "**Only while held**" → a Saturday action can never land after the feed has printed a new price.
- "**Only once it prints**" → the first-block action waits in the sequencer for the reopen.
- The sequencer accepts a bound at most 32 blocks ahead, so the keeper re-arms every minute; every check also lives in the contract.

Predicate	On Base Sepolia
`slot = held`	`0x6b17288f…49dd` · included next block
`slot ≠ held`	`0xbbd84631…b2fb` · included after the round changed
keeper, held	`0x982c7327…b695` · automatic protection
keeper, held	`0x6d45ddc9…40d9` · automatic, 89 min later
keeper, frozen	`0x9774913e…0352` · close-out, staged outage, 12 Oct
vault, frozen	`0x47ac5279…1d24` · ladder repay, same predicate

Measured: "not before", not "as soon as" (`docs/evidence/validity-transactions-sepolia.md`).

WHAT IS LIVE

Base Sepolia, on real B20 test stocks mirrored from mainnet.

What	What happened	Transaction
Weekend refusal	Sat 3 Oct: collateral added; a borrow attempt reverts `MarketClosed(WEEKEND)`	`0x01191a77…760b`
In-session draw	Tue 6 Oct, PRE: 5 USDC through the CreditDesk at $332.67, decision hash on chain	`0x661ebb1a…8bac`
Automatic protection	keeper pre-liquidated an in-band position at the live price: shadow LTV 60.12% → 58.00%	`0x982c7327…b695`
Reserve advance	protective repay on a live line: 0.5 USDC from the user's reserve + 0.2 USDC Reserve advance, settled with a 1% fee	`0x28c1ec44…08a4`
Local money	0.5 USDC → 0.46 EURCt inside the borrow flow, session-checked	`0x9090633c…5bd3`
x402	two keyless API calls paid $0.01 test USDC each	`0x8816d08f…c6ff`
Frozen feed	Mon 12 Oct, a staged feed outage: the keeper closed out an in-band position at the live price as a validity transaction; a raw liquidation reverted `FeedStale`; the vault repaid a second line from the user's reserve + a Reserve advance	`0x9774913e…0352`
Drills	stale feed, registry pause, issuer pause, scheduled multiplier, blocked receiver, keeper hold: each refused by name, repays still land	`0x84064c22…8429`
Webhooks	a hold and a release on a live line reached the partner HMAC-signed, 84 s and 170 s after the block	`0x5dd9d8df…42b7`
Dividend → loan	a dividend multiplier step swept into a desk line's debt: flash loan, repay first, sell only the dividend collateral	`0xa44b7867…f680`

Every Sepolia contract verified on sepolia.basescan.org (`deployments/base-sepolia.json`). The same footprint is on Base mainnet since 13 Oct: 50 transactions, 16 contracts verified on basescan.org, 33 of 33 checks (`deployments/base-mainnet.json`); its market is unseeded and nobody has opted in yet.

BASE MAINNET

Deployed 13 Oct; read live every minute, proven on forks, replayed through weekend gaps.

$1.73M

USDC borrowed on Aave V4's MAG7 stock spoke against $5.14M of tokenized-stock collateral (block 52,517,725, 12 Oct)

14

Morpho B20 markets, 16,315.76 USDC borrowed (12 Oct)

10

Coinbase stock feeds, sessions and registry flags, every minute

8

mainnet-fork suites: SessionGuard on the live Morpho, the Protector on a real Steakhouse borrower, Aave's Taker and Giver, the EURC pool, the replay, the Aave hub halt, a bad-debt probe, a real dividend

Replay, 7 Oct (fork block 52,295,895)

Gap at reopen	Raw venue: liquidated	Boundline: liquidated
–3%	0 of 5	0 of 5
–8%	1 of 5	0 of 5
–15%	2 of 5	0 of 5

Five positions: three as they stood, two real positions staged near their limit. Every raw liquidation was a staged position; the three as they stood stayed healthy even at –15%. Boundline acts on Saturday at the live price (pre-liquidation in the band, a repay above it). Evidence hash `0xa626f099…3c45`.

THE PARTNER SURFACE

A wallet integrates in a day and keeps its brand.

Partner API
Keys, quotes (capacity rounds down, charges round up, unknown data fails closed), obligations, decision timelines, policy versions, drills, evidence export. Public routes metered by **x402**. Route plans; HMAC-signed webhooks.

SDK + app
One `wallet_sendCalls` batch: authorise, approve, add collateral, draw, swap to local money. Preconfirmation in 200 ms. The Boundline app is partner #1.

Console + agents
Obligations, timelines, policy editor, drills, reconciliation, evidence with the hash recomputed in the page. An **MCP** server: session, positions, shadow health, quote, protect, route.

Live: app · console · API · agent

PUBLIC GOODS

MIT, deployed, usable by anyone without asking us.

Public good	What anyone can do with it
B20 Credit Observatory	read every Coinbase stock feed, session, registry flag, DEX gap and real Aave and Morpho position on Base, minute by minute, from an open JSON API
`MarketCalendar` + `@boundline/calendar`	ask onchain or in TypeScript which NYSE session any instant belongs to, holidays and early closes included
`SessionGate`, `SessionGuard`, `ShadowGuard`	list a Morpho market that refuses a frozen price; protect positions at a live price
The weekend dataset	two weekends recorded (9–11 Oct also captured live); rebuild any weekend's freeze and reopen from the chain (`scripts/reopen-evidence.py`, standard library only)
The replay harness	take real positions through weekend gaps on a fork and publish the result with a hash (first run 7 Oct; scheduled nightly in CI once the repository is public)

Fees on credit that already exists on Base.

MODEL

Platform fee per partner · 20–50 bps on originations · 10–25 bps a year on protected notional · x402-metered API · Reserve advance fees and the pre-liquidation incentive to capital providers.

TEAM

Farseen Shaikh, incoming MSc Data Science, TU Berlin · TODO: second developer, one line each

115 commits from 3 to 13 Oct, 264 Foundry + 898 TypeScript tests, eight mainnet-fork suites, eight services live, the footprint on Base mainnet.

THE ASK

$1M seed, 12 months

40% engineering · 20% legal and structure · 15% security (two audits) · 15% capital and partnerships (seed the Reserve) · 10% go-to-market

MILESTONES

Three design partners live on mainnet · $5M protected notional · the oracle adopted by a curator · a third venue

VALIDATION

$1.73M of stock-backed debt on Aave V4's stock spoke alone sits on a feed that freezes every weekend.

TODO: conversations, opt-ins

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Never liquidated while Wall Street is closed.

Observatory	https://observatory.boundline.xyz
App	https://app.boundline.xyz
Console	https://console.boundline.xyz
Site	https://boundline.xyz
Code	`TODO: public GitHub URL` · MIT
Every claim	`docs/submission/SUBMISSION.md` → evidence index; `deployments/base-sepolia.json`